MAYANK NARLA

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EDUCATION

University of California, Santa Cruz **Sep 2025 - Expected June 2028**
B.S., Computer Science (GPA: 3.81), Dean's Honors — Santa Cruz, CA

Foothill College **Jun 2024 - Jul 2026**
Dual Enrollment Program/Summer Quarters — Los Altos Hills, CA

EXPERIENCE

Software Engineering Intern — REC (My Recommendation) **Jun 2026 - Present**
Remote

- Developing with the development team on features across the REC app using Ruby on Rails and Flutter/Dart, contributing to backend services in an agile startup environment.
- Built an AI-powered content moderation pipeline using Ruby on Rails, Sidekiq, and the Gemini API to automate admin review of user submissions.

Researcher — LLM Logic Research – AIEA Lab **Mar 2026 - Present**
Santa Cruz, CA

- Built a neurosymbolic logical inference engine in Python combining LLMs with SWI-Prolog, translating natural language queries into formal Prolog via a LangGraph pipeline.
- Implemented RAG, relevancy judging, and chain-of-thought refinement using LangChain and LangGraph.
- Ran GPU training jobs on the Nautilus distributed computing cluster using Kubernetes, training a CNN on CIFAR-10.

Computer Vision Engineer — UCSC Rocket Team **Sep 2025 - Feb 2026**
Santa Cruz, CA

- Collaborated and engineered a homography matrix and real-time video stitching pipeline in Python and OpenCV to map onboard camera footage for landing localization and recovery tracking.

TECHNICAL PROJECTS

Paddock Scout (https://github.com/mnarla/Paddock-Scout) **May 2026 - Jun 2026**

- Built a regulation-aware F1 race prediction system utilizing a Random Forest classifier with temporal weighting, featuring automated CI/CD pipelines for real-time telemetry ingestion.
- Engineered an autonomous Python-based web intelligence agent that scrapes technical news (via DuckDuckGo) to detect and validate team performance upgrades, dynamically updating model features with no API costs.

WFM Sell-Timing Advisor (https://github.com/mnarla/wfmarket-advisor) **Jul 2026 - Aug 2026**

- Built a multi-node LangGraph pipeline (trend, vault, patch, and synthesis nodes) that ingests live Warframe Market pricing and in-game state data to generate SELL/HOLD trading recommendations.
- Designed a SQLite-based conditional caching layer with independent tracking per data source, minimizing redundant API calls while keeping price, vault, and patch data current.

Culler (https://github.com/mnarla/culler) **Aug 2026 - Sep 2026**

- Developed a local agentic curation engine using Qwen3 8B to predict track skips by processing Spotify and Last.fm datasets, implementing an automated self-calibration loop for heuristic rule refinement through error-based clustering.
- Architected a SQLite-backed system to track rule creation and integrated full chain-of-thought reasoning, successfully improving prediction grounding by prioritizing logical consistency over raw speed.

Formova (https://github.com/mnarla/formova) **May 2025 - Jun 2025**

- Developed a real-time exercise form analysis tool in Python using OpenCV and MediaPipe, and built an AI chatbot with Gemini API to deliver instant fitness feedback.

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, SQL, Ruby, C, HTML/CSS, Assembly (RISC-V)

Frameworks: Ruby on Rails, Flutter, LangChain, LangGraph, OpenCV, MediaPipe, NumPy, PyTorch, TensorFlow, Scikit-learn, Pandas, OpenAI API, Gemini API, Large Language Models, Generative AI

ML/AI Concepts: Machine Learning, Deep Learning, Neural Networks, CNNs, Retrieval-Augmented Generation (RAG), Computer Vision, NLP, Statistics

Tools: SWI-Prolog, Janus, Kubernetes, Nautilus Cluster, AWS, SQLite, PostgreSQL, Git, Linux/Unix Shell, Raspberry Pi, VS Code, Google Cloud APIs

CERTIFICATIONS & BADGES

- Anthropic Academy: Introduction to MCP (Model Context Protocol)
- AWS Cloud Quest: Cloud Practitioner, Solutions Architect (Training Badges)